



Modul : Decision Tree Learning (DTL)

Variable (Attribute) Types

**Source: DataMining Concepts and Techniques
by Jiawei Han, Micheline Kamber, Jian Pei**

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**Pembelajaran Mesin
(Machine Learning)**



Numeric

Quantitative
(measurable
quantity) ? Integer
or Real Values

Interval Scaled
(equal-size units,
have order)

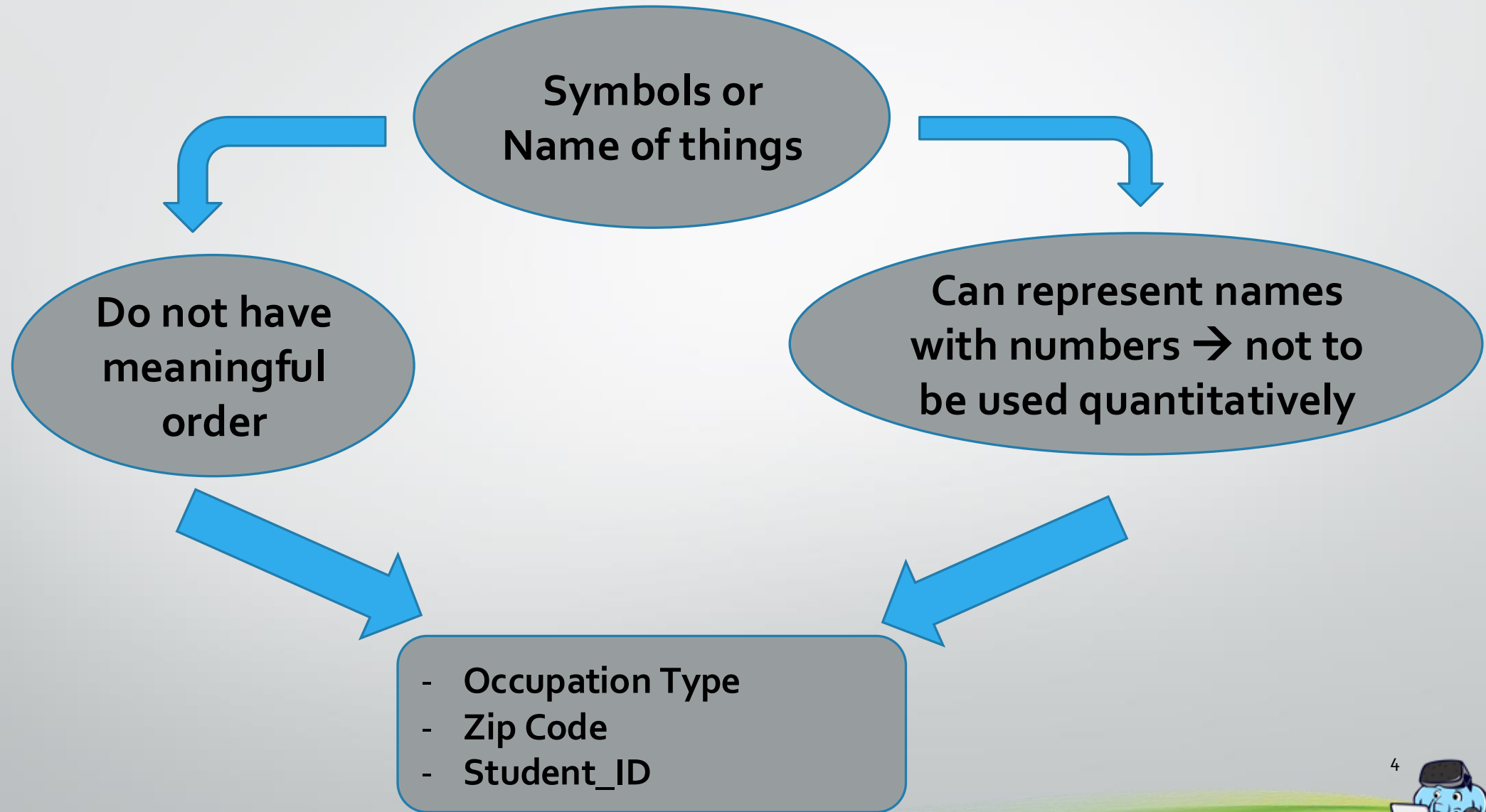
- Temperature
- Calendar dates

Ratio-Scaled
(inherent zero
point, a value can
be multiple of
another value)

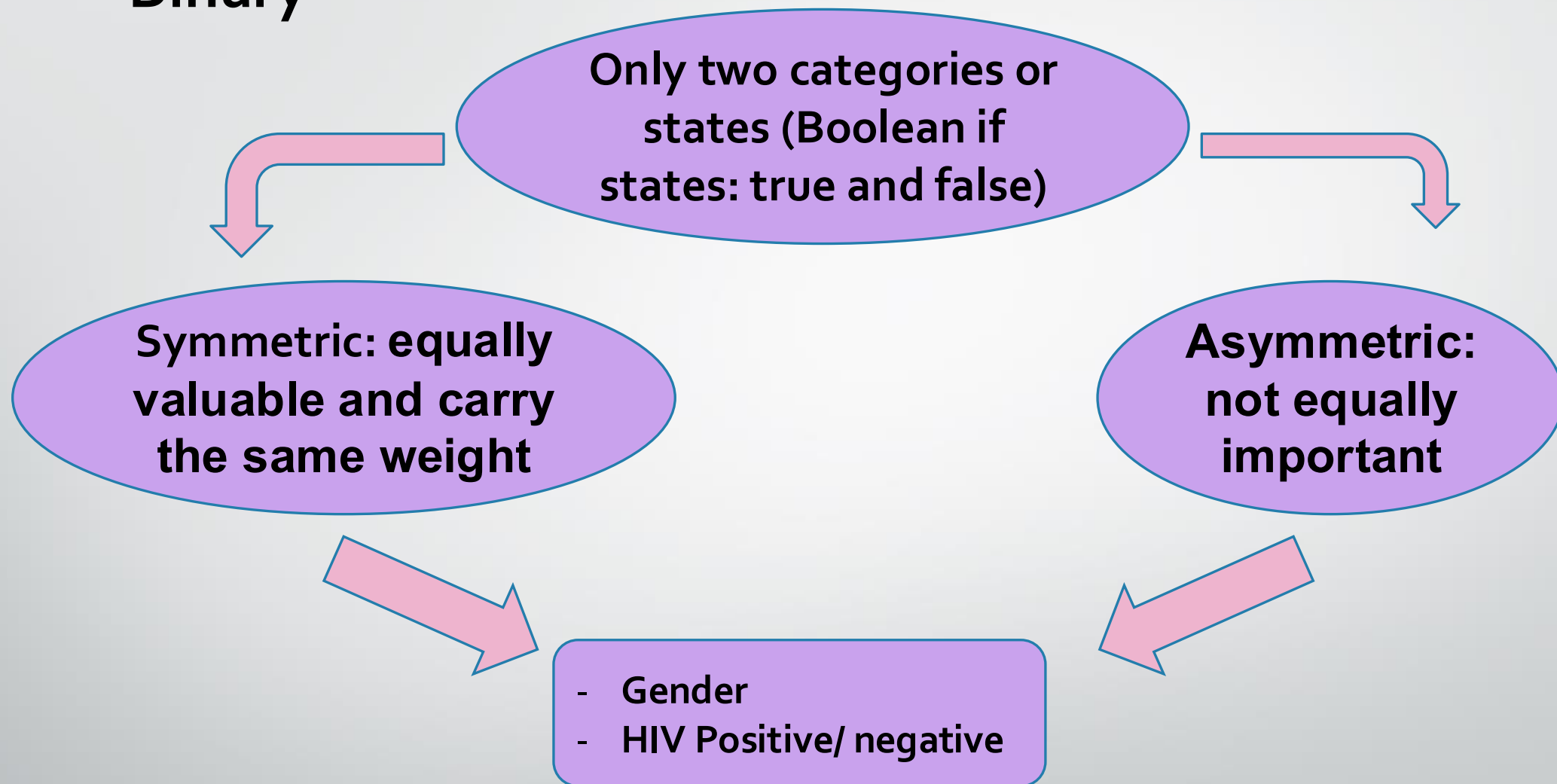
- Years of experience
- Weight
- Number of words



Nominal/ Categorical



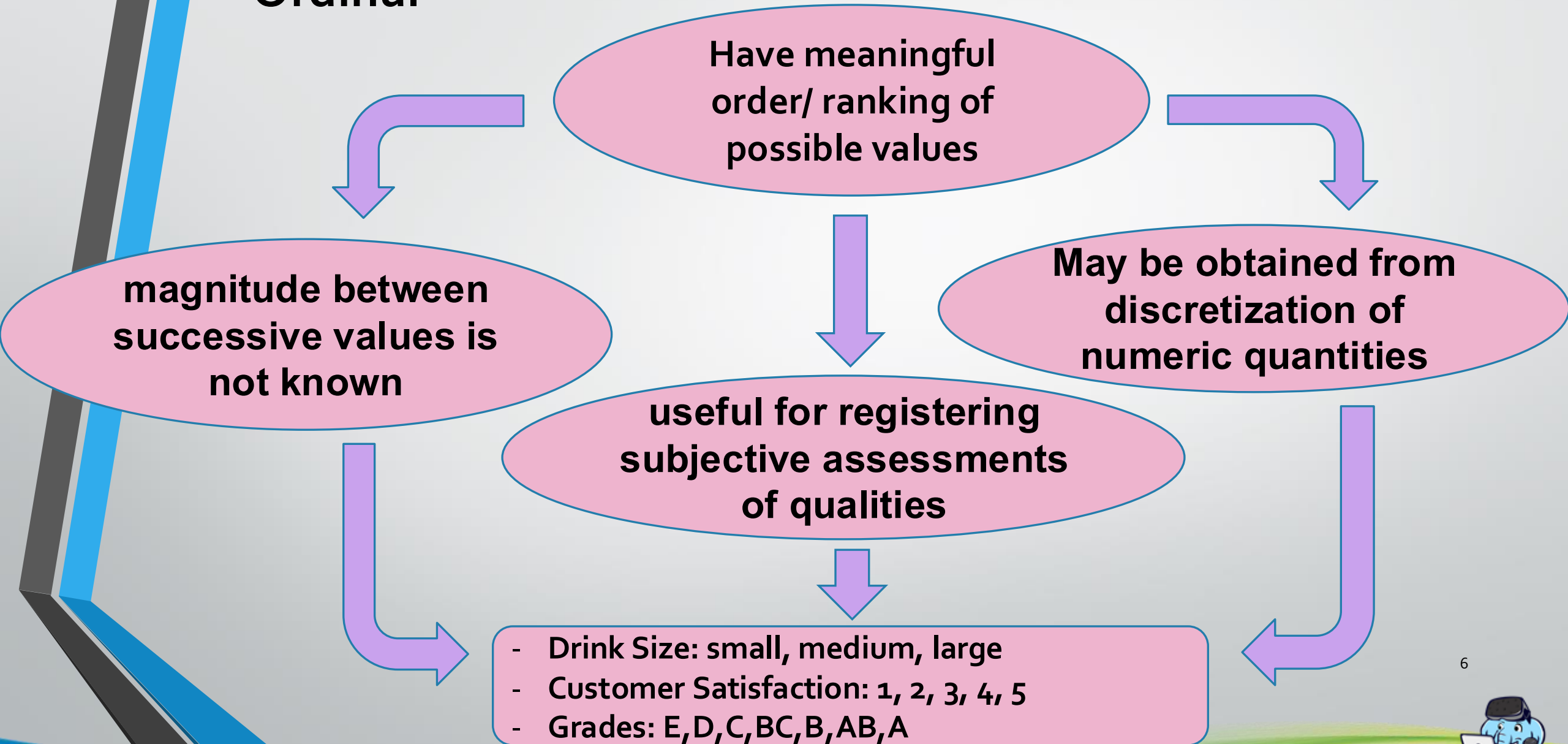
Binary



1	red,	green,	blue
2	1,	0,	0
3	0,	1,	0
4	0,	0,	1



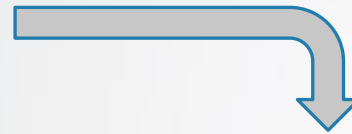
Ordinal



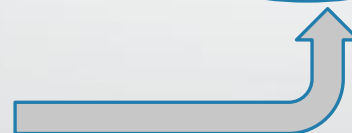
Discrete vs Continuous

has a finite set of values:
Drink size, Age, Medical
test,

has a countably infinite
set of values:
Customer ID, Zip code



Discrete



$\neq$

Continuous



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Modul : Decision Tree Learning (DTL)

What, Why, and When

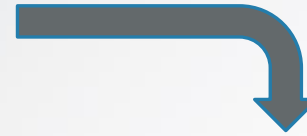
Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**

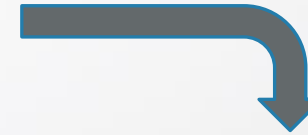


WHAT is DTL

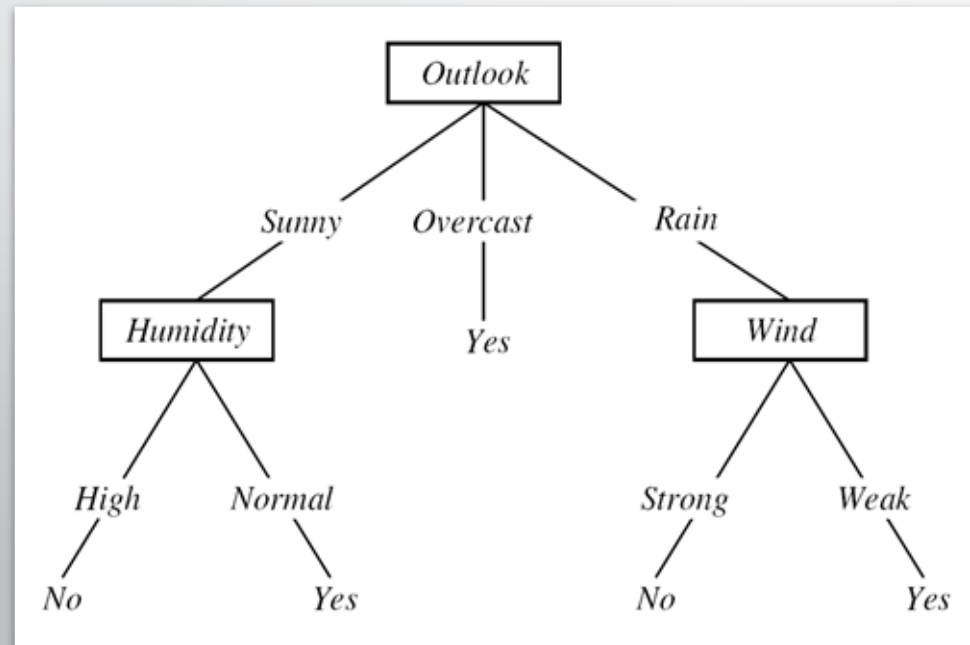
**Method for
approximating
discrete-valued target
functions**



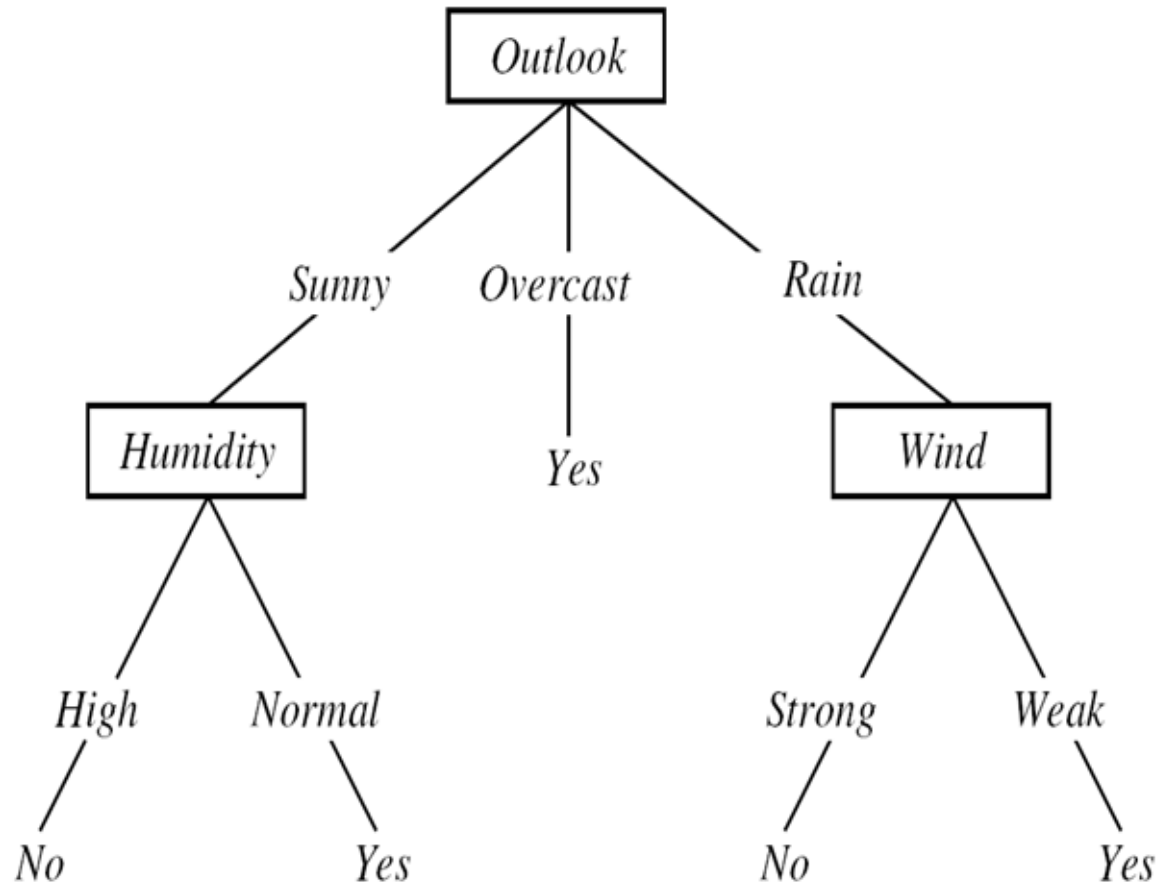
**Represented by
Decision Tree**



**Can be
represented as set
of if-then rules**



DTL Representation



- Each internal node represents test of an attributes
- Branch descending from the node corresponds to one possible value
- Leaf/ terminal nodes represents classification result

Represent a disjunction of conjunctions of constraints on the attribute values of instances

$(\text{outlook}=\text{sunny} \wedge \text{humidity}=\text{normal})$
 $\vee (\text{outlook}=\text{overcast})$
 $\vee (\text{outlook}=\text{rain} \wedge \text{wind}=\text{weak})$



WHY DTL

**Method for
approximating
discrete-valued target
functions**

**Represented by
Decision Tree or
If-Then Rules**

Robust to Noisy Data

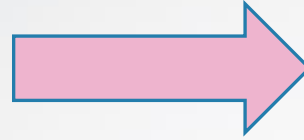
**Capable of learning
disjunctive expression**

- Popular of inductive inference algorithms
- Have been successfully applied to a broad range of tasks: learning to diagnose medical cases, learning to assess credit risk of loan applicants



When we use DTL → Appropriate Problems for DTL

Instances are represented
by attribute-value pairs



For continuous attribute:
discretization

The target function has
discrete output values

Disjunctive descriptions
may be required

The training data may
contain errors

The training data may
contain missing attribute
values



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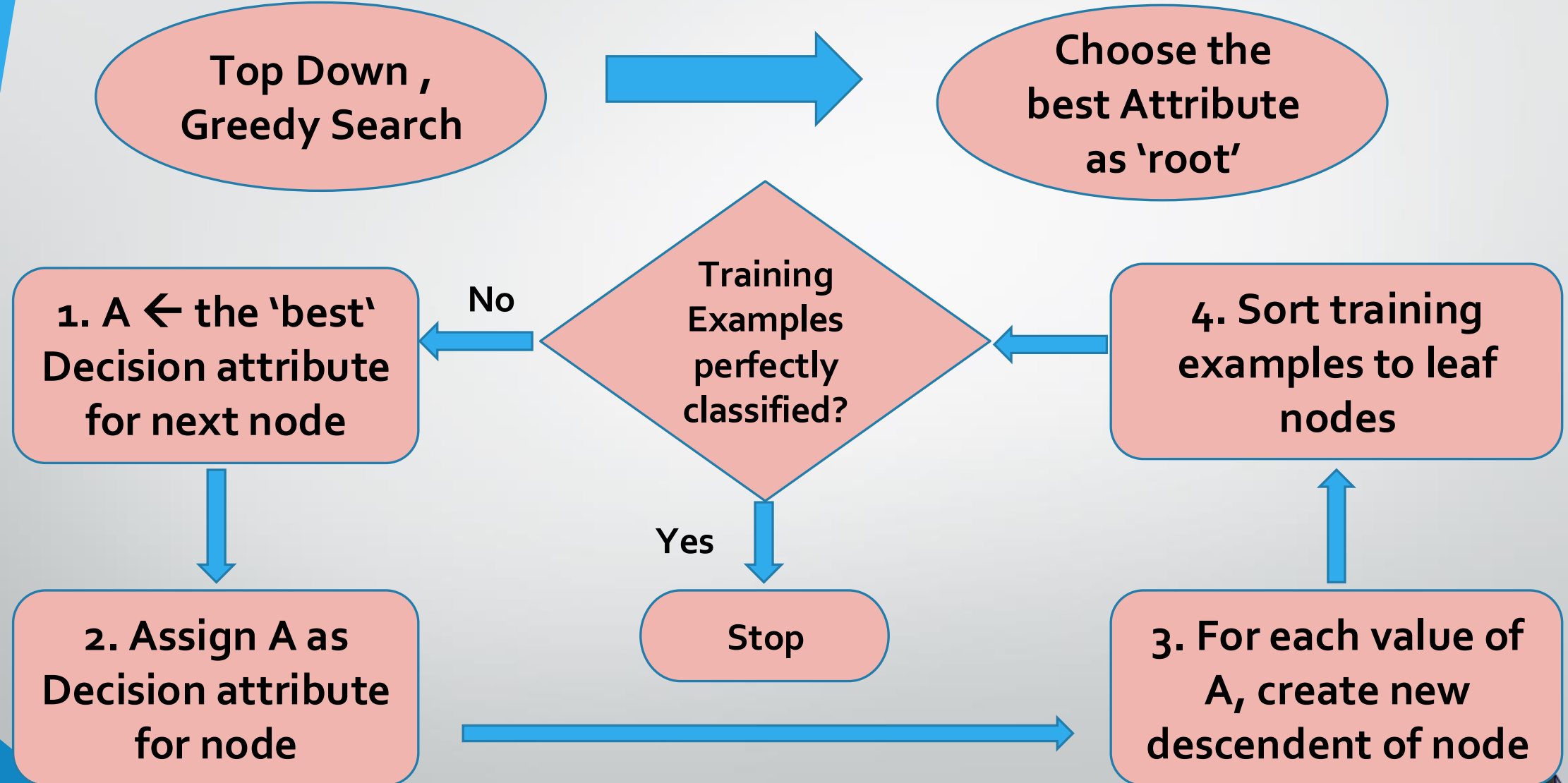
Basic DTL Algorithm (ID3)

Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**

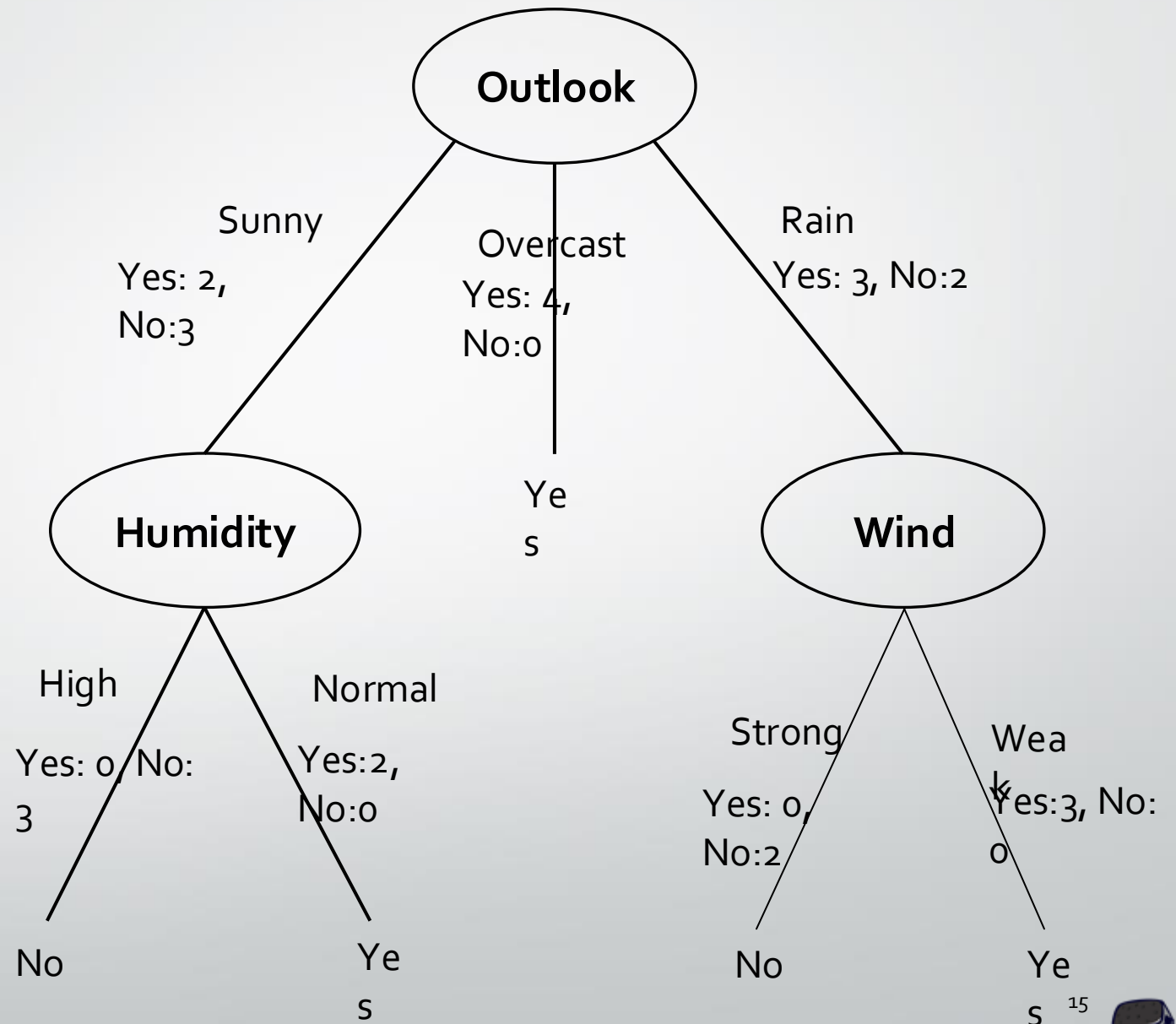


ID₃ Algorithm



Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



Decision Tree Learning (Russel & Norvig, 2021)

function DECISION-TREE-LEARNING(*examples*, *attributes*, *parent_examples*) **returns**
a tree

if *examples* is empty **then return** PLURALITY-VALUE(*parent_examples*)
else if all *examples* have the same classification **then return** the classification
else if *attributes* is empty **then return** PLURALITY-VALUE(*examples*)
else

$A \leftarrow \operatorname{argmax}_{a \in \text{attributes}} \text{IMPORTANCE}(a, \text{examples})$

tree $\leftarrow$ a new decision tree with root test *A*

for each value v_k of *A* **do**

$\text{exs} \leftarrow \{e : e \in \text{examples} \text{ and } e.A = v_k\}$

subtree $\leftarrow$ DECISION-TREE-LEARNING(*exs*, *attributes* – *A*, *examples*)

add a branch to *tree* with label (*A* = v_k) and subtree *subtree*

return *tree*

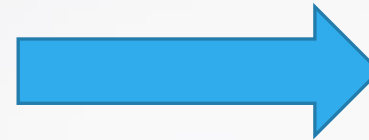
Information gain
Gain ratio

|*A*|=2: binary tree
|*A*| min 2: n-ary tree

The function PLURALITY-VALUE selects the most common output value among a set of examples, breaking ties randomly.

Best Attribute

“Ideally” can classify the training examples into subsets, which has the same class



Information Gain



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Information Gain

Source: Machine Learning, Tom Mitchell

**Pembelajaran Mesin
(Machine Learning)**



Entropy (Information Content)

Measurement in
Information Theory
□ impurity of an
arbitrary collection
of samples

S = set of training examples
Entropy: the
minimum number of bits of
information needed to
encode the classification of
an arbitrary member of S

$$Entropy(S) \equiv \sum_{i=1}^c -p_i \log_2 p_i$$

- S : set of training examples
- c : number of classes
- p_i : proportion of S belonging to class i



Entropy for S with 2 values/ classes

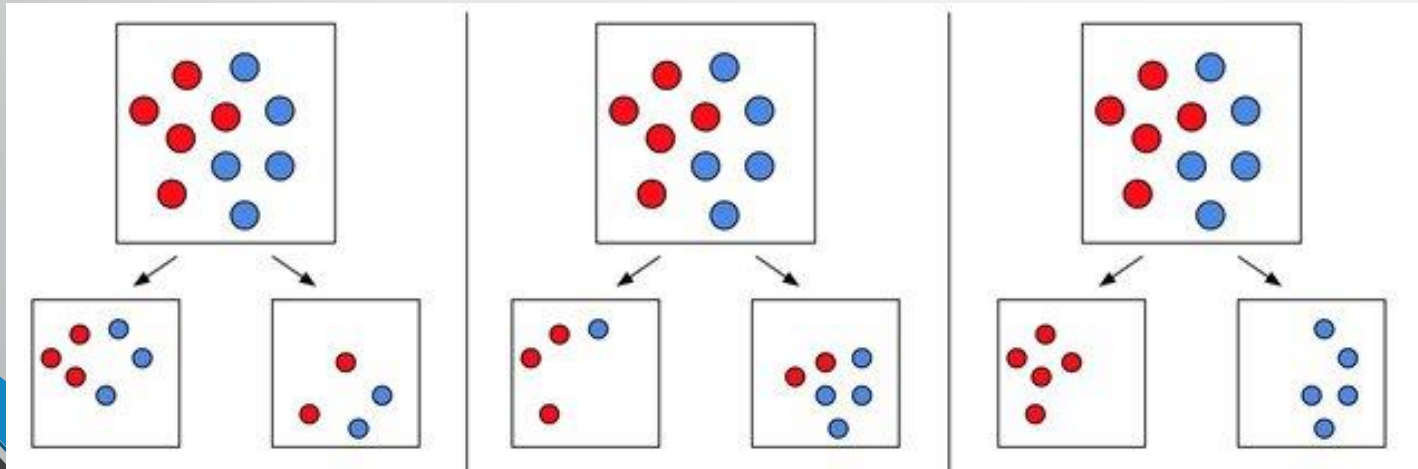
S contains positive examples and negative examples

$$Entropy(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus}$$

Entropy 0: all examples belong to a class (no surprises, no message need be sent)

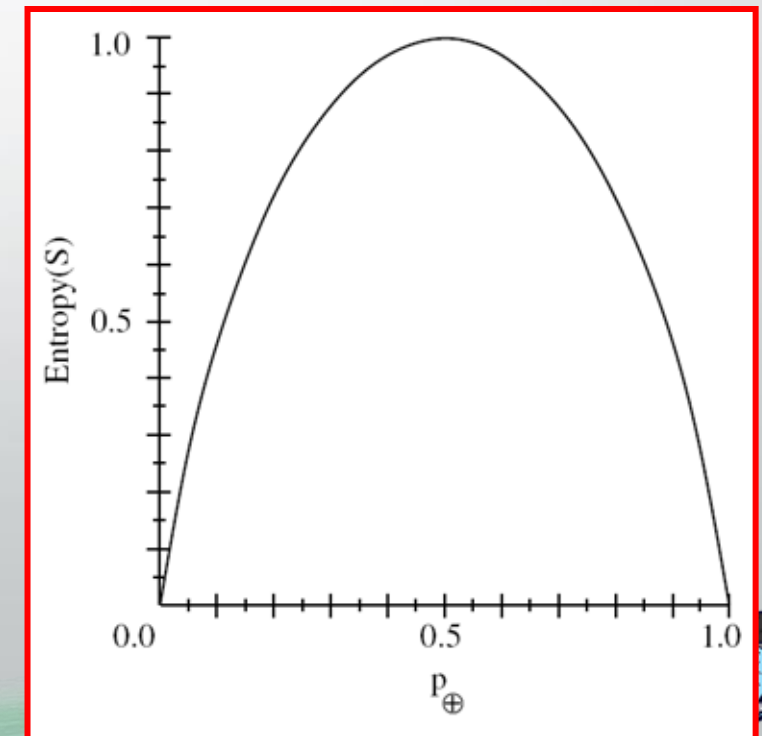
Entropy 1: $p_{+} = p_{-}$
(1 bit is required to indicate the class of the drawn example)

$$I(5/10, 5/10) = 1$$



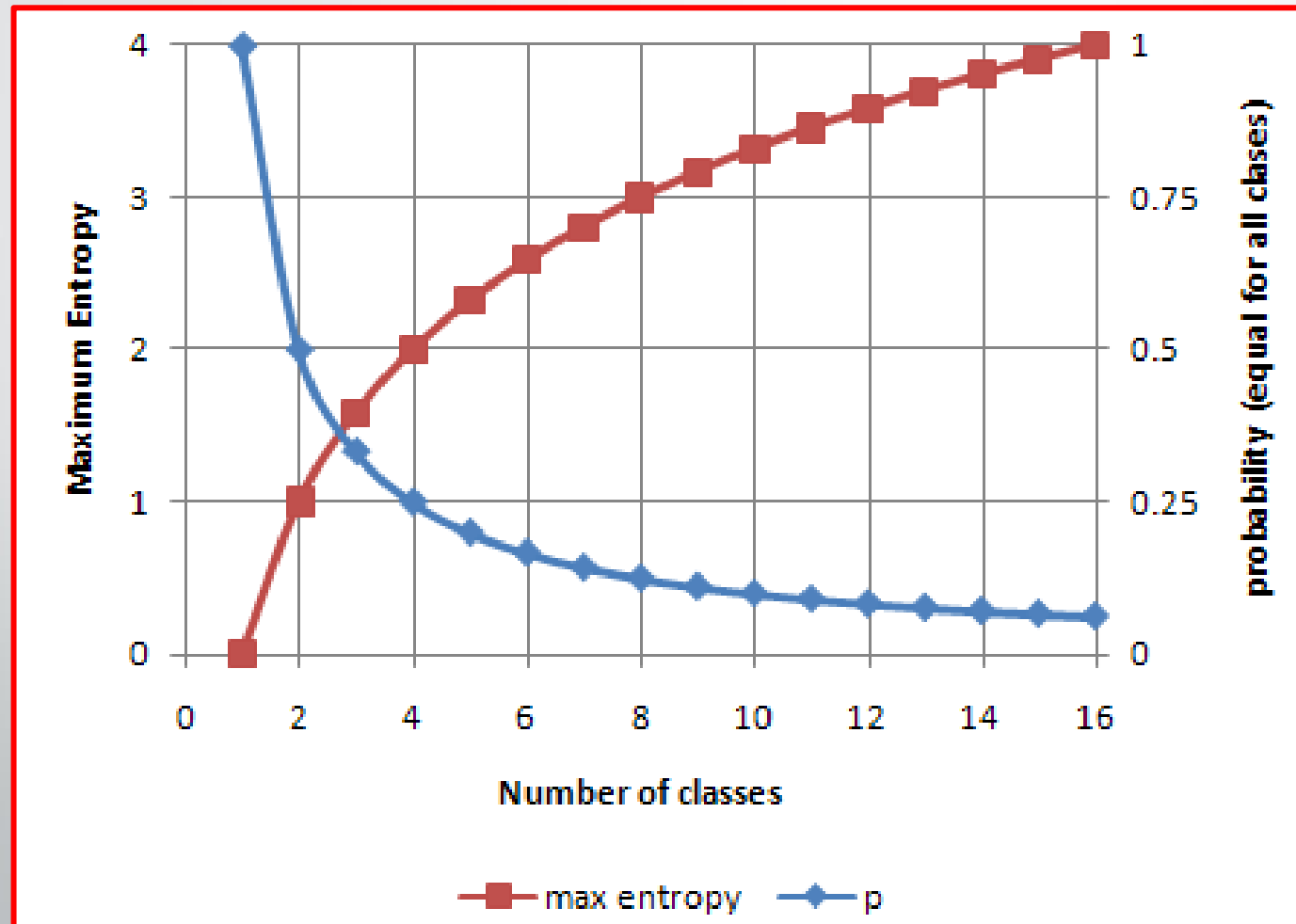
$$I(3/6, 3/6) = 1 \quad I(2/4, 2/4) = 1 \quad I(3/4, 1/4) = 0.81 \quad I(5/5, 0/5) = 0 \quad I(0/5, 5/5) = 0$$

$$I(2/6, 4/6) = 0.92$$



Entropy for S with n values/ classes

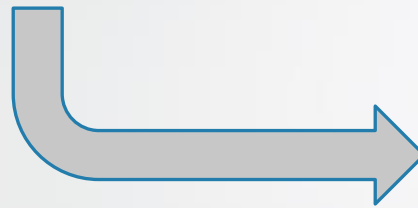
Target attribute has n values: entropy can be as large as $\log_2 n$



Information Gain

Gain(S,A) = expected reduction of entropy due to sorting A

S: set of training examples; A: an attribute



Find A which has maximum Gain(S,A)

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$



Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

$$\text{Gain}(S, \text{Outlook}) = \text{Entropy}(S) - \sum_{v \in \text{value of Outlook}} \frac{S_v}{S} \text{Entropy}(S_v)$$

$$= 0.940 - \left[\frac{5}{14} \text{Entropy}(\text{Sunny}) + \frac{4}{14} \text{Entropy}(\text{Overcast}) + \frac{5}{14} \text{Entropy}(\text{Rain}) \right]$$

$$= 0.940 - \left[\left(\frac{5}{14} \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right) \right) + \left(\frac{4}{14} \left(-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right) \right) + \left(\frac{5}{14} \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) \right) \right]$$

$$= 0.246$$



Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

$$\text{Gain}(S, \text{Humidity}) = \text{Entropy}(S) - \sum_{v \in \text{value of Humidity}} \frac{S_v}{S} \text{Entropy}(S_v)$$

$$= 0.940 - \left[\frac{7}{14} \text{Entropy}(\text{High}) + \frac{7}{14} \text{Entropy}(\text{Normal}) \right]$$

$$= 0.940 - \left[\left(\frac{7}{14} \left(-\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \right) \right) + \left(\frac{7}{14} \left(-\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \right) \right) \right]$$

$$= 0.151$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Outlook}),$
 $\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

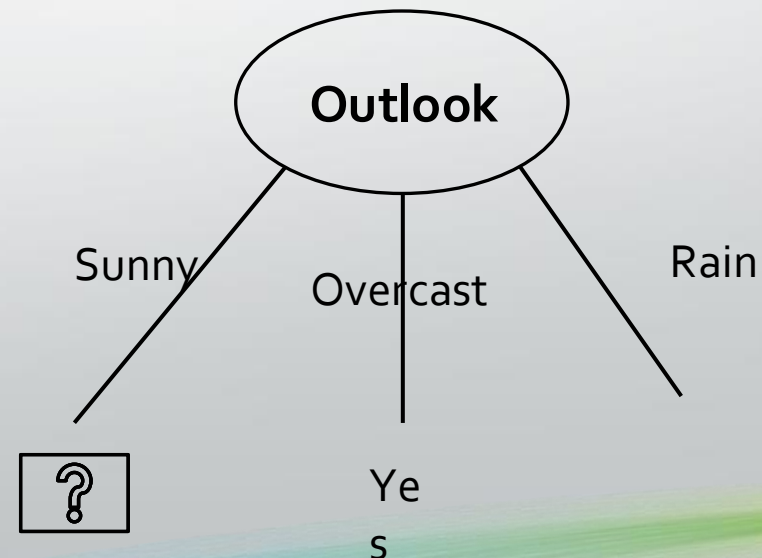
$$\text{Gain}(S, \text{Outlook}) = 0.246$$

$$\text{Gain}(S, \text{Humidity}) = 0.151$$

$$\text{Gain}(S, \text{Temperature}) = 0.029$$

$$\text{Gain}(S, \text{Wind}) = 0.048$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



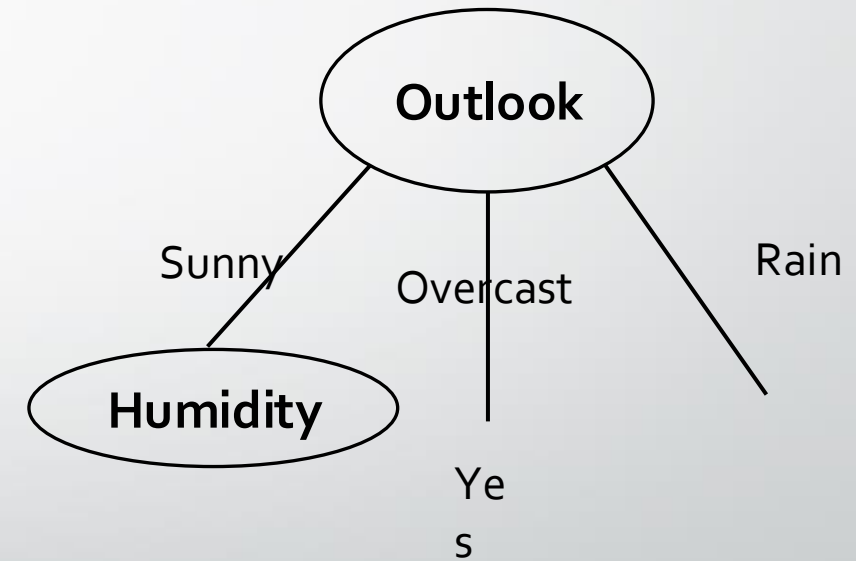
Example

Find the 'best' Attribute:
 $\max(\text{Gain}(S, \text{Temperature}),$
 $\text{Gain}(S, \text{Humidity}),$
 $\text{Gain}(S, \text{Wind}))$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes

$$\text{Entropy}(S) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$$

$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= \text{Entropy}(S) - \sum_{v \in \text{value of Humidity}} \frac{S_v}{S} \text{Entropy}(S_v) \\ &= 0.971 - \frac{3}{5} \text{Entropy}(\text{High}) - \frac{2}{5} \text{Entropy}(\text{Normal}) \\ &= 0.971 - \left[\left(\frac{3}{5} \left(-\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} \right) + \left(\frac{2}{5} \left(-\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \right) \right) \right) \right] \\ &= 0.971 \end{aligned}$$



$$\text{Gain}(S, \text{Humidity}) = 0.971$$

$$\text{Gain}(S, \text{Temperature}) = 0.571$$

$$\text{Gain}(S, \text{Wind}) = 0.020$$



Exercise (Class: Aktivitas)

No	Deadline?	Ada Hangout?	Malas?	Aktivitas
1	Urgent	ya	Ya	Kumpul-kumpul
2	Urgent	Tidak	Ya	Belajar
3	Dekat	ya	Ya	Kumpul-kumpul
4	Tidak ada	Ya	Tidak	Kumpul-kumpul
5	Tidak ada	Tidak	Ya	Jalan-jalan ke mall
6	Tidak ada	Ya	Tidak	Kumpul-kumpul
7	Dekat	Tidak	Tidak	Belajar
8	Dekat	Tidak	Ya	Nonton TV
9	Dekat	Ya	Ya	Kumpul-kumpul
10	Urgent	Tidak	Tidak	Belajar





THANK YOU



